



 JURALCO

JURALCO EDGETEC INFINITY® BALUSTRADE SYSTEM

ISSUE 2-25 v1

Producer statement cover sheet for structural glass balustrade systems

One cover sheet must be provided for each balustrade system

PART 1: TO BE COMPLETED BY THE DESIGN PROFESSIONAL / MANUFACTURER SUPPLYING THE GLASS SYSTEM

Cover sheet issued on behalf of:	Royal Glass
In respect of (system type):	Edgetec Infinity® Balustrade
Supplied to:	Auckland Council
Client / designers name:	Josh Harrison
Description of work: <i>E.g. New dwelling</i>	12mm thick Glass Balustrade installation for External Balcony area using Channel System
Address of property:	83 Speights Road, St Heiliers, Auckland 1071

PART 2: CONSTRUCTION DETAILS *(tick boxes that apply)*

Deck / balcony / stairs construction:	☒ Timber	☐ Precast	☐ Concrete	☐ Steel	☐ <i>please specify</i>	
Location of deck / balcony / stairs:	☐ Internal	☒ External	☐ <i>please specify</i>			
Deck / balcony / stairs:	☒ New	☐ Existing	Cladding:	☒ Direct fixed	☐ Cavity	☐ N/A

PART 3: GLASS DETAILS *(tick boxes that apply)*

Type:	☒ Toughened	☐ Toughened laminated	☐ Toughened laminated with stiff interlayer	
Thickness:	12mm (mm)		Width of pane:	0.5-1.8 (m)
Height above floor:	(m)		Height above fixing:	(m)
Note: If barrier extends more than 1.2m above fixing point, test reports must be provided				
Supplier:	Royal Glass			

PART 4: BARRIER DETAILS

System name:	Edgetec Infinity Balustrade
Drawings (ref page number)	
Engineers name:	Lautrec Technology Group
Hardware (type):	Aluminium Clamps

Drawings must show the type of hardware being used to support the glass (i.e. channel, discs, etc.)

PART 5: SAFETY FEATURES (tick all boxes that apply)☒ Continuous interlinking rail (show how fixed to building and each pane of glass)☐ Continuous interlinking top cap (show how fixed to building and each pane of glass)☐ Clamps, brackets or point fixings (show how fixed to building and each pane of glass)☐ Framed with structural sealant min. 2 edges (show how fixed to building and each pane of glass)☐ Glass with stiff interlayer☐ Two or three glass edges supported by continuous clamps☐ Structural post to fix interlinking rail to deck☐ Other (specify)

Engineers name: Lautrec Technology Group

Drawings (ref page number)

Comments:

PART 6: COVER SHEET COMPLETED BY

Name: Roxy Huang

Signature: 

Date: 14/05/2025

Contact number: 0800 769 254

Email: info@royalglass.co.nz

NAME: Josh Harrison

ADDRESS:

DATE: 14/05/2025

CONSENT AUTHORITY: Auckland Council



THIS PS1 DOES NOT CONSTITUTE A SITE SPECIFIC REVIEW BY THE ENGINEERING COMPANY BELOW AND IS ONLY APPLICABLE FOR STANDARD FIXINGS AND DESIGN PARAMETERS AS DETAILED IN THE RELEVANT MANUAL



association of
consulting and
engineering

Building Code Clause(s): B1, F2, F4, F9

PRODUCER STATEMENT – PS1 – DESIGN

ISSUED BY: Lautrec Technology Group Limited
(Design Firm)

TO: Juralco Aluminium Building Products Limited
(Owner/Developer)

TO BE SUPPLIED TO: All Building Consent Authorities in NZ (Auckland Council Author Number: 1385)
(Building Consent Authority)

IN RESPECT OF: Juralco Edgetec® Infinity Balustrade System
(Description of Building Work)

AT: N/A (all locations within NZ Wind Zone)
(Address)

Town/City: LOT DP SO
(Address)

We have been engaged by the owner/developer referred to above to provide:

Specific Engineering Design Structural Components Only
(Extent of Engagement)

services in respect of the requirements of Clause(s): B1, F2, F4, F9 of the Building Code for:

☐ All or ☒ Part only (as specified in the attachment to this statement), of the proposed building work.

The design carried out by us has been prepared in accordance with:

☒ Compliance Documents issued by the Ministry of Business, Innovation & Employment
B1/VM1; A5/NZS 1170:2021; A5/NZS 1664.1:1997;
NZS AS 1720.1:2022; NZS 3101:2006; NZS 3404:1997;
NZS 4223.3:2016; NZS 4223.4:2008; NZS 8500:2016
Of (verification method/acceptable solution)

☐ Alternative solution as per the attached schedule.

The proposed building work covered by this producer statement is described on the report titled:

"PS1 Report - Juralco Edgetec® Infinity Balustrade System_27.02.25" together with product manual "Juralco Edgetec® Infinity Balustrade System Issue 2-25 v1"

together with the specification, and other documents set out in the schedule attached to this statement.

On behalf of the Design Firm, and subject to:

- (i) Site verification of the following design assumptions: assumed adequate support structure by others
(ii) All proprietary products meeting their performance specification requirements;

I believe on reasonable grounds that a) the building, if constructed in accordance with the drawings, specifications, and other documents provided or listed in the attached schedule, will comply with the relevant provisions of the Building Code and that b), the persons who have undertaken the design have the necessary competency to do so. I also recommend the following level of construction monitoring/observation:

☐ CM1 ☐ CM2 ☐ CM3 ☐ CM4 ☐ CM5 (Engineering Categories) or ☒ as per agreement with owner/developer (Architectural)

I, Kevin Brown am: ☒ CPEng # 140404
(Name of Design Professional)

I am a member of: ☒ Engineering New Zealand and hold the following qualifications: BE, CMEngNZ, CPEng, IntPE(NZ), MBA.

The Design Firm issuing this statement holds a current policy of Professional Indemnity Insurance no less than \$200,000*.

The Design Firm is a member of ACE New Zealand: ☒

SIGNED BY: Kevin Brown (Signature)
(Name of Design Professional)

ON BEHALF OF: Lautrec Technology Group Limited Date: 27/02/2025
(Design Firm)

Note: This statement shall only be relied upon by the Building Consent Authority named above. Liability under this statement accrues to the Design Firm only. The total maximum amount of damages payable arising from this statement and all other statements provided to the Building Consent Authority in relation to this building work, whether in contract, tort or otherwise (including negligence), is limited to the sum of \$200,000*.

This form is to accompany Form 2 of the Building (Forms) Regulations 2004 for the application of a Building Consent.
THIS FORM AND ITS CONDITIONS ARE COPYRIGHT TO ACE NEW ZEALAND AND ENGINEERING NEW ZEALAND

27/02/2025

To the Building Official,

B2 COMPLIANCE – JURALCO EDGETEC® INFINITY BALUSTRADE SYSTEM

at occupancy categories A, B, E, and C3; at any location in New Zealand that falls within the scope of this PS1 - see supporting PS1 report for scope

We have been asked to provide a PS1 for Clause B2 of the Building Code - Structural Durability

We are not able to provide this because there is no effective verification method for B2 contained within the New Zealand Building Code.

As these systems can be installed in a variety of settings, including internal and exposed environments, it is not deemed practical to specify durability requirements for the sub structure. Timber treatments, mild steel corrosion protection coatings, and concrete and masonry covers are therefore up to the building designer to specify in accordance with the relevant recognised standards.

However, we can confirm that for the structural elements shown in the attached documentation:

Material	Means of compliance	Details
Juralco Edgetec® Infinity Balustrade System - Aluminium and Stainless Steel	Alternative Solution	Stainless steel components conform to SS316. Aluminium extrusions, components, and hardware conform to 6060 T5. Non-structural cladding has a durability requirement not less than 50 years for this project. Note that the industry proven life of Aluminium, Stainless Steel, and Glass would satisfy a 50-year design life requirement. We note that this is on a time to first maintenance basis.
Connections - Stainless Steel fixings	Alternative Solution	All bolt and screw fixings within Juralco Edgetec® Infinity Balustrade System shall be minimum SS316. Non-structural fixings that are difficult to replace has a durability requirement not less than 50 years for this project. Note that the industry proven life of Stainless Steel would satisfy a 50-year design life requirement.

It is assumed that these structural elements are fixed to adequate structures by others.

Minor tea staining may occur in coastal environments. Refer to Juralco Edgetec® Infinity Balustrade System manual issue 2-25 v1 for the supplier's maintenance requirements.

Yours faithfully,

Managing Director
Kevin Brown
BE, CMEngNZ, CPEng, IntPE(NZ), MBA



Juralco Edgetec Infinity® Balustrade System

Juralco Aluminium Building Products Ltd designs and distributes specialist aluminium joinery systems through a national network of franchised fabricators and agents. For more than 25 years we have been at the forefront of specialist aluminium door and window products suitable for New Zealand joinery and building methods. Our comprehensive product range includes security and insect screens, balustrades and gates, shutters and awnings, shower screens, wardrobe doors and organisers and internal doors.

The Juralco Edgetec Infinity® Balustrade system is designed for Frameless Glass, from 12mm to 17.52mm, either Top or Faced fixed and for Residential or Commercial use.

An Interlinking Top Rail (depending on Glass type) must be used.

The system is extremely versatile and can be made in a range of configurations to suit most modern architectural requirements.

The Infinity Semi Frameless Glass system features heavy duty internal clamps at regular intervals all covered by continuous cover extrusions front and back, giving a streamlined minimal look. For Top or Face fixing.

- Juralco Edgetec Infinity® Balustrade System
- Glass Panels from 12mm Toughened Safety Glass to 17.5mm SentryGlas®
- Tested to NZ standard NZS4203 and NZS1170
- Conforms to NZS 4223.3.2016
- Top Interlinking Rail to conform to NZS 4223.3.2016
- Clamps spaced at 400mm - 500mm centres depending on the application and Glass type
- Simple installation. Allows horizontal and vertical glass adjustment.



Top Fix System + Interlinking Top Rail



Top Fix System + SentryGlas. Interlinking Top rail not required



Top Fix System + Interlinking Top Rail



Face Fix System + Interlinking Top Rail

Juralco Edgetec Infinity® Balustrade Patent #NZ 630364
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Juralco Edgetec Infinity® Balustrade System

Complies With AS/NZS 1170:2002, NZS 4223.3.2016, NZ Building Code B1, B2, F2, F4 and F9

**For Domestic and Residential Occupancy types A, A Other and C3 and
for Commercial Occupancy Types B, E, A Other and C3
Occupancy Types as per AS/NZ 1170.1.2002. Not suitable for Commercial C1/C2, C5 and D applications**

Code	Type of Occupancy for part of the building or structure	Specific Uses	Glass
A	Domestic and Residential activities	All areas within or serving exclusively one dwelling including stairs, landings etc, but excluding external balconies and edges of roofs.	Residential, 12mm Toughened Glass, 15.2 mm Laminated or 13.52mm SentryGlas®
B, E	Offices and work areas not included elsewhere including storage areas.	Light access stairs and gangways not more than 600mm wide Fixed platforms, walkways, stairways and ladders for access Areas not susceptible to overcrowding in office and institutional buildings; also industrial and storage building.	Commercial, 15mm Toughened Glass, 17.2 mm Laminated or 17.52mm SentryGlas®
A Other, C3	Areas without obstacles for moving people and not susceptible to over crowding	Stairs, landings, external balconies, edges of roofs etc.	Residential or Commercial as detailed above

Note 1 All for 12mm or 15mm Toughened, 15.2mm or 17.2mm Laminated and 13.52mm or 17.52mm SentryGlas® . All edges polished.

Note 2 Juralco Balustrade Systems building code compliance documentation requires all balustrade installations are to be completed in accordance with the requirements of our authorised installer certification.

Note 3 Frameless Glass Balustrades must conform to NZS 4223.3.2016
See individual Layout pages for conformance details

masterspec partner
Section 4852JB

Note 4 The Dulux powder coating warranty period is conditional upon the Balustrade being maintained in accordance with the Dulux 'Care and Maintenance Instructions'. See Page 5 for warnings concerning Coastal conditions.
Contact your balustrade installer for a copy of the Care and Maintenance procedure.

Index

Heading	Pages	Description. Use the Bookmarks  List to jump to selected pages
Specifications	4	Juralco standard specification sheet and Powder coating recommendations
All Sections Below for Face Fix only		
Configurations	5 - 7	Shows typical elevation layouts for Residential and Commercial for all Glass types
General	8	Shows Clamp Cross section and all details
Components	9 - 12	Shows all Components and Extrusions
Face Fix Options	13	Shows Approved Face fix Options into Timber
Mountings	14 - 26	Shows Mounting details, all for Face Fixed. Timber (p13-22), Steel (p23-24) and Concrete (p25)
Installation	27	Face Fix recommended Installation guide.
All Sections below for Top Fix only		
Configurations	28 - 29	Shows typical elevation layouts for Residential and Commercial for all glass types
General	30	Shows Clamp Cross section and all details
Components	31 - 33	Shows all Components and Extrusions
Mountings	34 - 38	Shows Mounting details, all for Top fixed. Timber (p32-33), Steel (p34-35) and Concrete (p36)
Installation	39	Top Fix recommended Installation Guide
Glass Top Edge - Safety		
Glass Panels. Top edge. Safety	40 - 49	Shows all Interlinking Rail options (5 x Total). Swivel connectors and End connections
	50	Shows Glass stiffener brackets, Fixed and Adjustable angles
End Brackets	51 - 53	Shows Interlinking Rail End bracket attachments to Posts and Structures
Rail mounted to Glass Panel	54	Shows Interlinking Rail Side mounted to a Glass Panel and all End brackets
Joiners and End Plates	55 - 56	Shows Interlinking and Handrail Joiners and End Plates
Surface Care	57 - 59	Instructions for the care of Glass, Powder Coated and Stainless Steel surfaces



Juralco Edgetec Infinity® Balustrade System - Specifications, Powder Coating

Juralco Aluminium Building Products Ltd (JABP) Specifications for Juralco Edgetec Infinity® Balustrade System

1. Scope

- This specification details the documents the Juralco Edgetec Infinity® Balustrade System refers to in relation to the New Zealand Building Code, the manufacturer's documents, products used in the System, requirements in relation to fixing and surface finishing.

2. NZBC Compliance

- The Juralco Edgetec Infinity® Balustrade System has been reviewed by Lautrec Technology Group Ltd to demonstrate compliance with the structural requirements of the New Zealand Building Code and NZS 1170 : 2002 occupancy A, B, E, A Other and C3, NZS 3604 Low, Medium, High, Very High and Extra High Wind Zones, to a maximum ULS wind load of 2.5kPa
- The Structural Engineering design includes the requirements of B1 Structure, B2 Durability, F2 Hazardous material and F4 Safety from falling, all from the Building Code.
- Verification Method B1 / VM1, B2/AS1, F4 / AS1
- All glass used in the Juralco Edgetec Infinity® Balustrade System must conform to AS/NZS 2208. Complies with NZS 4223.3.2016
- Separation of dissimilar materials (as relates to B2 compliance) have been reviewed.
For other combinations refer to NZS 3604:2011 Section 2.3.3 Separation and Section 4 Durability

3. Manufacturer's Documents

- The Juralco Edgetec Infinity® Balustrade System manual details all extrusions and components used for the fabrication and installation/fixing of the system.
- A Producer Statement 1(Design) is available.
Copies of the above documents are available from:
Juralco Aluminium Building Products Ltd
48 Bruce McLaren Rd, Henderson, Auckland
Phone 09 478 8018 Fax 09 478 7883 Email specify@juralco.co.nz
- Any deviation from the standard fabrication or installation/fixing must be accompanied by a site specific PS1 with site specific calculations and drawings

4. Products

- Only extrusions, components and hardware supplied by or specified by JABP may be used in the Juralco Edgetec Infinity® Balustrade System
- Aluminium extrusions, components and hardware – unless specified are manufactured to 6060 T5 specifications
- Stainless Steel components, hardware, fixings – all components to 304 or 316 grade
- Glass - all glass used in the Juralco Edgetec Infinity® Balustrade System must conform to the specifications as listed in the Juralco Edgetec Infinity® Balustrade System manual with each panel conforming to AS/NZS 2208 as confirmed by the Safety Stamp detailing the manufacturer's description and licence number.

5. Surface Finishing

- Juralco Aluminium Building Products Ltd is a Dulux Registered Applicator site, registration number 2101.
JABP uses only Dulux branded powder coating materials
- Dulux Duralloy® powder coating systems are suitable for properties greater than 100m from high tide level
AAMA 2603 performance. Residential buildings, 3 levels max. Warranty 10 yrs
- Dulux Duralloy Plus® powder coating systems are suitable for properties greater than 10m from high tide level.
AAMA 2603 performance. Residential and Light commercial buildings, 3 levels max. Warranty 15 yrs
- Dulux Duratec® powder coating systems are suitable for properties greater than 10m from high tide level
AAMA2603 and 2604 performance. All Residential and Commercial buildings. Warranty 25 yrs

6. Installation and Fixing

- The Juralco Edgetec Infinity® Balustrade System must only be installed in accordance with the Juralco Edgetec Infinity® Balustrade System manual
- Any deviation from that specified in the Juralco Edgetec Infinity® Balustrade System manual must only be in accordance with the site specific PS1 with site specific calculations and drawings listing the non standard details
- The Juralco Edgetec Infinity® Balustrade System must only be fabricated/installed by a Juralco approved fabricator
- Upon completion of the installation the fabricator must supply the Council with a PS3 (Construction)

Important information - Powder Coating systems.

Powdercoat Systems The new standard Dulux powder coating system used by Juralco is Duralloy Plus®. Also Duralloy® and Duratec®. All as per specs above. Juralco Powder coated prices are for Duralloy Plus® and Duralloy® (same pricing). Duratec® prices on application.

Attachment to structures A PVC Tape or similar material spacer must be used to separate powder coated aluminium items from all concrete and steel structures. Failure to do so can lead to the chemicals in the structure affecting the powder coating, leading to corrosion.

Swimming Pools The chlorinated water in swimming pools can cause the deterioration of powder coated surfaces, leading to corrosion of the underlying surface. It is recommended that Powder coated surfaces be 1200mm min from a pool.

Care The Dulux powder coating warranty period is conditional upon the surface being maintained in accordance with the Dulux 'Care and Maintenance Instructions'. Download from Dulux or refer to the back page of this manual.



Juralco Edgetec Infinity® Balustrade System

Typical Layouts - Top Fix

Edgetec Infinity® Glass Clamps Top Fix + Interlinking Rail

Glass must have a minimum strength of 100Mpa. All edges polished

Residential & Domestic only Occupancy types A, A Other and C3

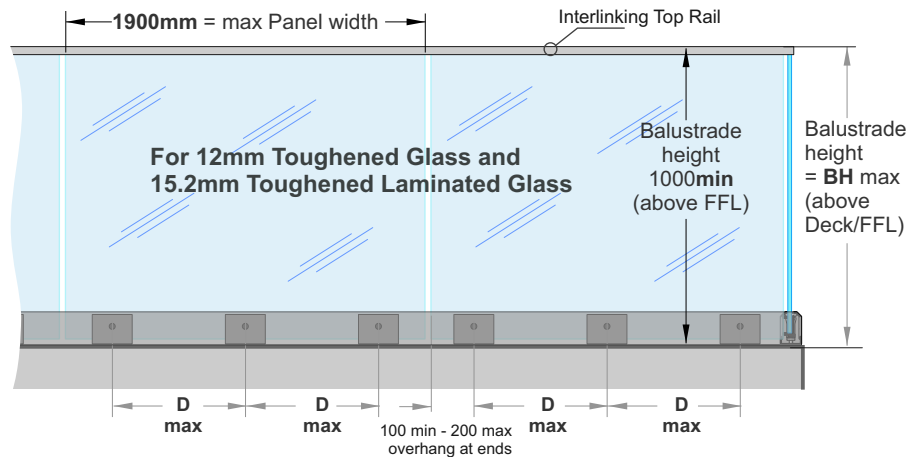
For 12mm Toughened Glass

- VHWZ** **EHWZ**
 - D max 500mm. - D max 500mm.
 - BH max 1200mm - BH max 1000mm

For 15.2mm Toughened Laminated Glass

- D max 500mm.
 - BH max 1150mm

Note: See individual Mounting pages for construction options



Max Tension load per Face fixing = 20kN

Exceeds the wind loading for all Wind Zones up to **and Including Extra High Wind Zone** as set out in NZS 3604:2011

Refer to the Interlinking Top Rail page for conformance to NZS 4223.3.2016.

Edgetec Infinity® Glass Clamps Top Fix + Interlinking Rail

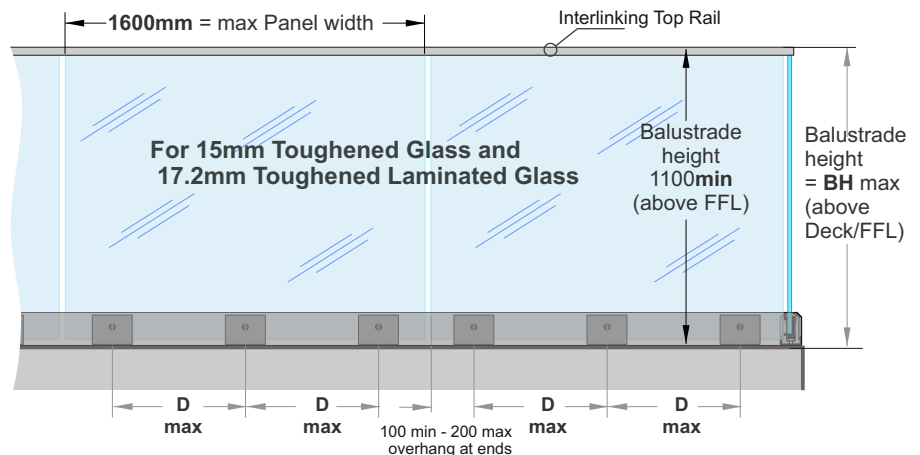
Glass must have a minimum strength of 100Mpa. All edges polished

Commercial Occupancy types B, E, and C3 only

For 15mm Toughened Glass and 17.2mm Toughened Laminated Glass

- D max 400mm
 - BH max 1300mm

Note: See individual Mounting pages for construction options



Max Tension load per Face fixing = 23kN

Exceeds the wind loading for all Wind Zones up to **and Including Extra High Wind Zone** as set out in NZS 3604:2011

Refer to the Interlinking Top Rail page for conformance to NZS 4223.3.2016.

Edgetec Infinity® Glass Clamps Top Fix

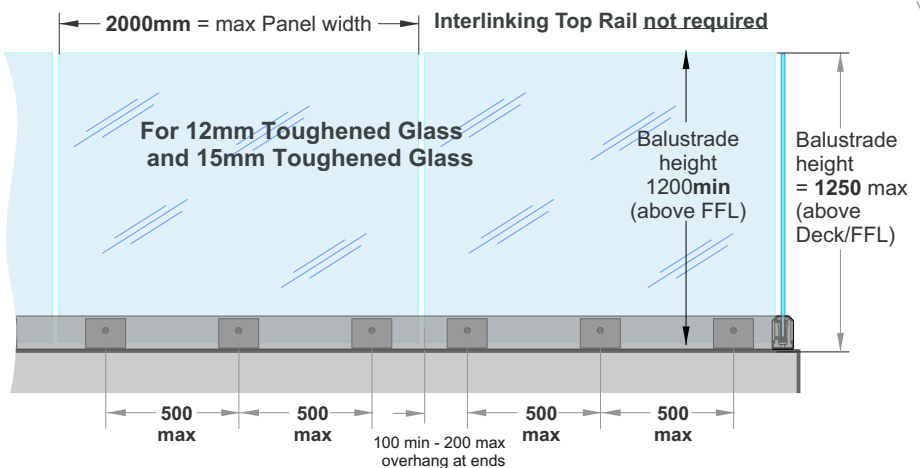
POOL FENCING only

Glass must have a minimum strength of 100Mpa. All edges polished

Max Tension load per Face fixing = 27kN

Applies to Swimming Pools as of Jan 2017, complies with the Building Code clause F9 and section 162C of the Building Act.

Applies to Pool Fences not protecting a fall of 1.0m or more



12mm Toughened - Up to and including **Very High Wind Zone.**

15mm Toughened - Up to and including **Extra High Wind Zone**

Juralco Edgetec Infinity® Balustrade System

Typical Layouts - Top Fix

Edgetec Infinity® Glass Clamps Top Fix + Stiffener Brackets

Glass must have a minimum strength of 100Mpa. All edges polished

Residential & Domestic only Occupancy types A, A Other and C3

For 15.2mm Toughened Laminated Glass

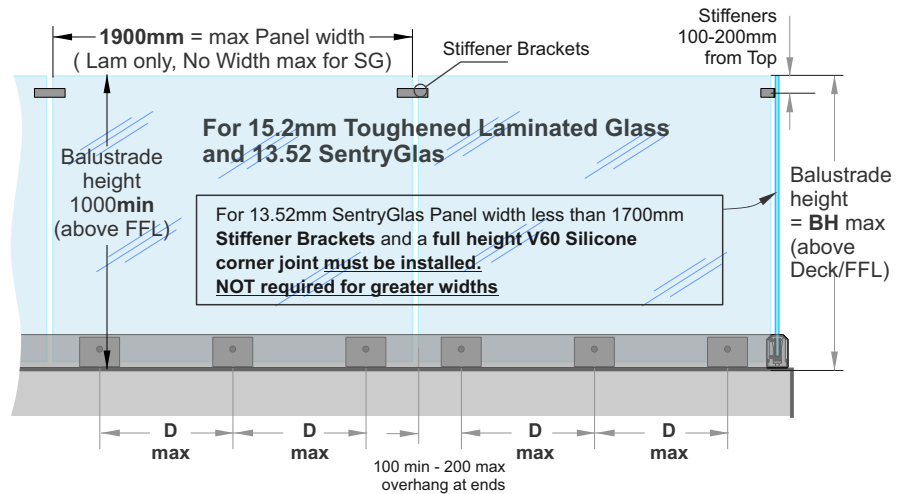
- D max 500mm.
- BH max 1150mm

For 13.52mm SentryGlas

- D max 350mm.
- BH max 1050mm

Note: See individual Mounting pages for construction options

Max Tension load per Face fixing = 20kN



Exceeds the wind loading for all Wind Zones up to **and Including Very High Wind Zone** as set out in NZS 3604:2011

Refer to the Stiffener Bracket pages for conformance to NZS 4223.3.2016.

Edgetec Infinity® Glass Clamps Top Fix + Stiffener Brackets

Glass must have a minimum strength of 100Mpa
All edges polished

Commercial Occupancy types B, E, and C3 only

For 17.2mm Toughened Laminated Glass

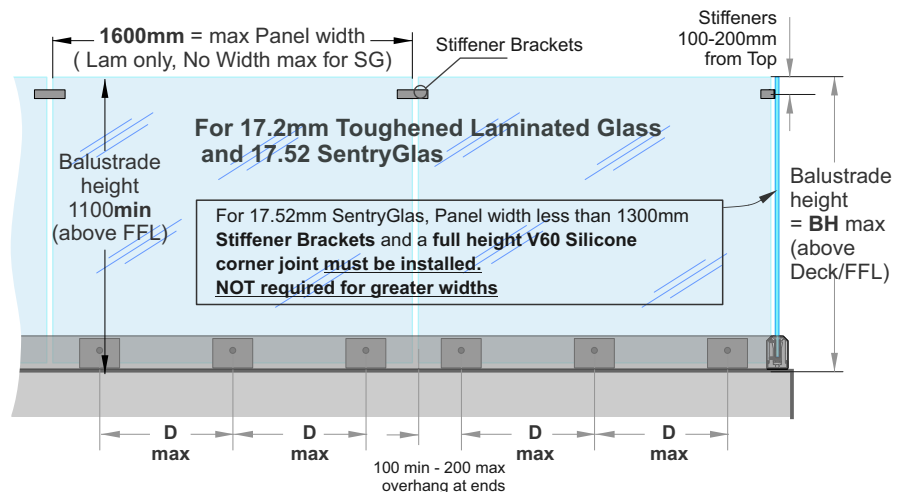
- D max 400mm
- BH max 1300mm

For 17.52mm SentryGlas

- D max 350mm
- BH max 1200mm

Note: See individual Mounting pages for construction options

Max Tension load per Face fixing = 23kN



Exceeds the wind loading for all Wind Zones up to **and Including Extra High Wind Zone** as set out in NZS 3604:2011

Refer to the Stiffener Bracket pages for conformance to NZS 4223.3.2016.

SentryGlas® Glass Layers and Thickness Orientation

Glass Thickness (mm)	Inner Layer of Glass thickness (mm) Deckside	Interlayer thickness(mm) and Type	Outer Layer Glass thickness (mm)
13.52	6	1.52 SentryGlas®	6
17.52	8	1.52 SentryGlas®	8

Refers to previous page. Laminated Glass Layers and Thickness Orientation

Glass Thickness (mm)	Inner Layer of Glass thickness (mm) Deckside	Interlayer thickness(mm) and Type	Outer Layer Glass thickness (mm)
15.2	8	1.2EVA	6
17.2	8	1.2EVA	8

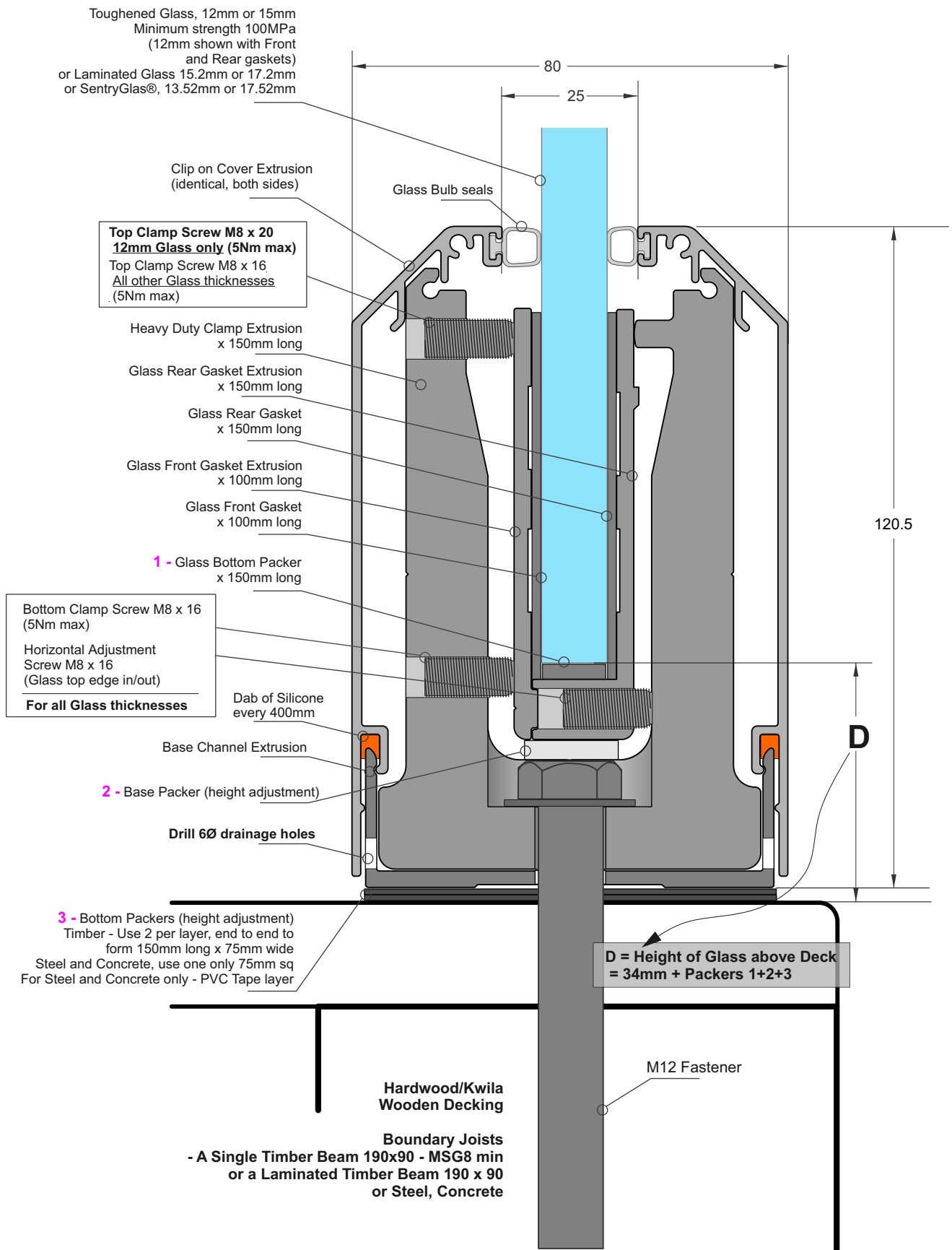


Juralco Edgetec Infinity® Balustrade System

Top Fix General

Edgetec Infinity®
Glass Clamp - Top Fix
(12mm Glass Shown)

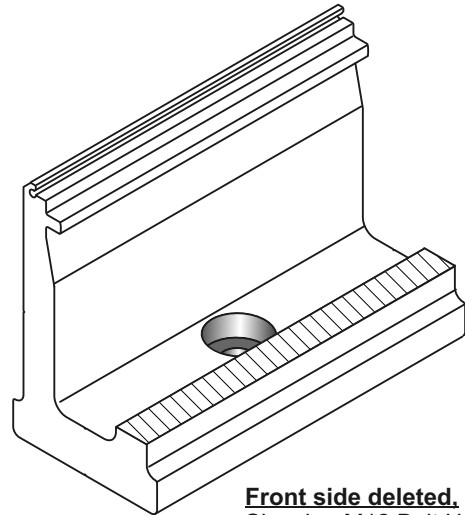
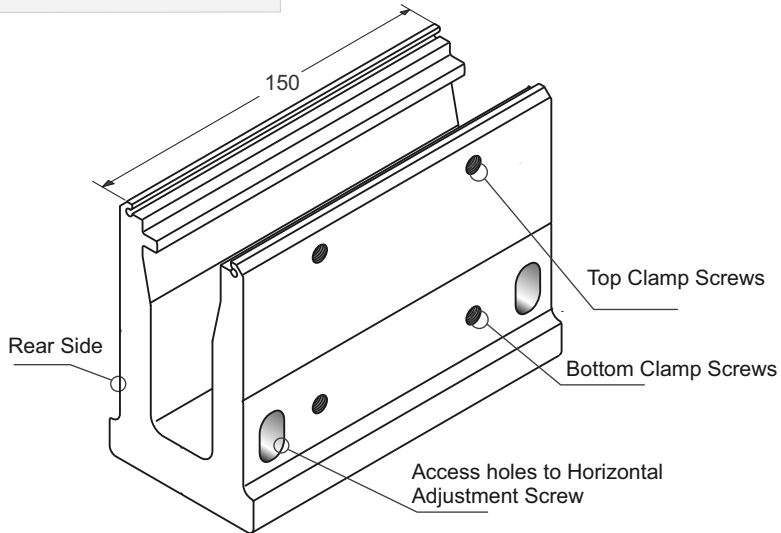
The Infinity Balustrade Clamp comes as a kit;
Clamp Extrusion, Front and Rear Gasket Extrusions
Gaskets, Glass bottom Packer and all adjusting screws.
(M12 Fastener not included)



Elevation showing the Main Features

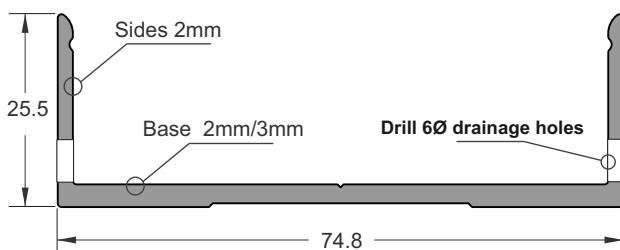
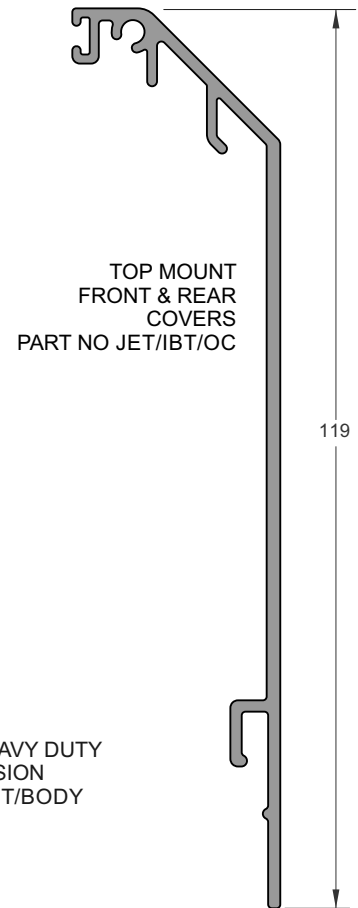
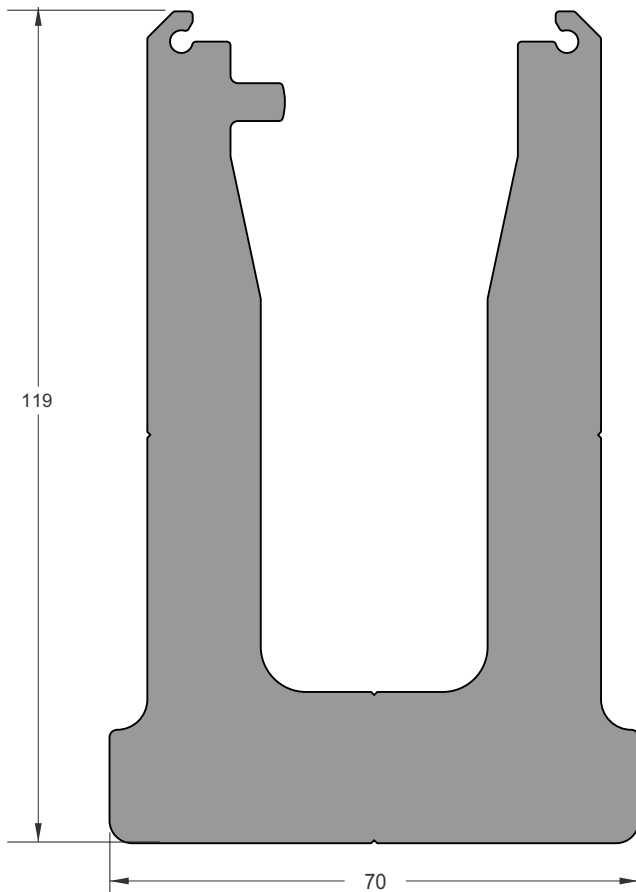
Juralco Edgetec Infinity® Balustrade System Top Fix Components

Heavy Duty Clamp Extrusion
Part No JET/IBT/BODY



Cover and Base Extrusion

Front and Back Cover Extrusions Identical



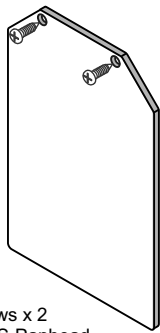
TOP MOUNT
BOTTOM CHANNEL
EXTRUSION
PART NO JET/IBT/BC



Juralco Edgetec Infinity® Balustrade System

Top Fix Components

Top Mount End Plate Part No JET/IBT/EP

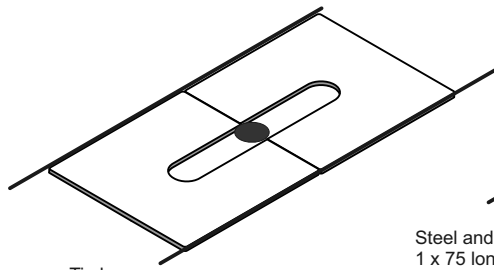


Attach Screws x 2
No 8 x 25 SS Panhead

Bottom Spacer Plates

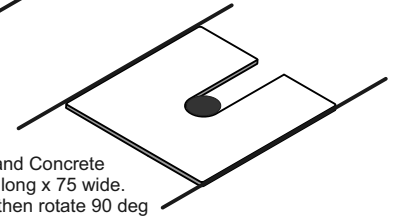
Packers (height adjustment)

- Timber - Use 2 per layer, end to end to form 150mm long x 75mm wide
- Steel and Concrete, use one only 75mm sq



Timber
2 x 75 long x 75 wide

1mm thick Plate - Part No JET/IBT/BSP1

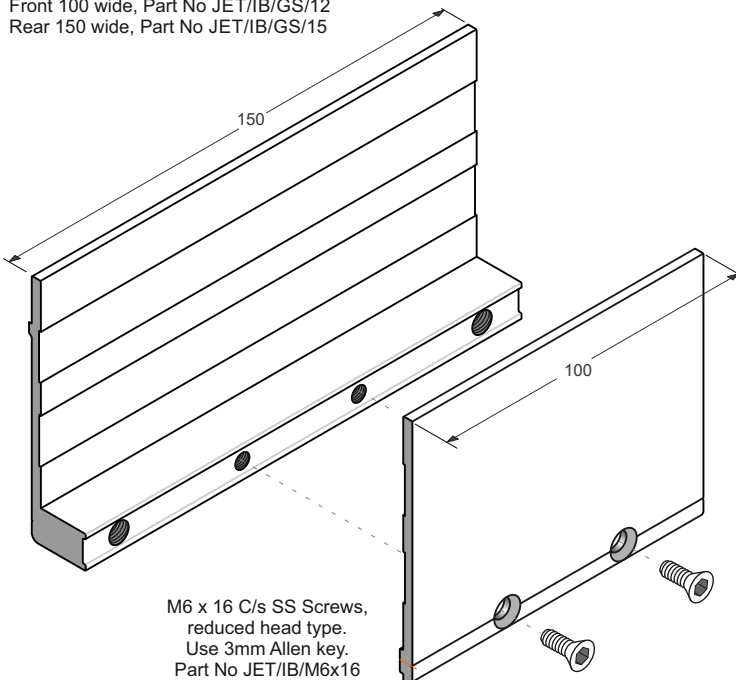


Steel and Concrete
1 x 75 long x 75 wide.
Insert then rotate 90 deg

5mm thick Plate - Part No JET/IBT/BSP5

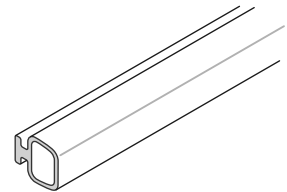
Glass Gasket Extrusions Front and Rear

Front 100 wide, Part No JET/IB/GS/12
Rear 150 wide, Part No JET/IB/GS/15

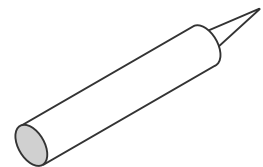


M6 x 16 C/s SS Screws,
reduced head type.
Use 3mm Allen key.
Part No JET/IB/M6x16

Glass Bulb Seal Part No JET/IB/CVRBLB250

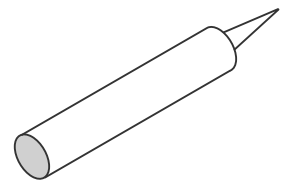


SIKA Supergrip 2hr Part No JEC SUPERGRIP



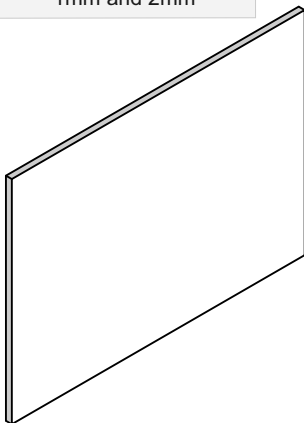
For All Coachscrews fixings

Rhodorsil V60 Clear Silicone Part No H/RTV419098



Construction Silicone

Glass Gaskets 1mm and 2mm



Gasket Schedule

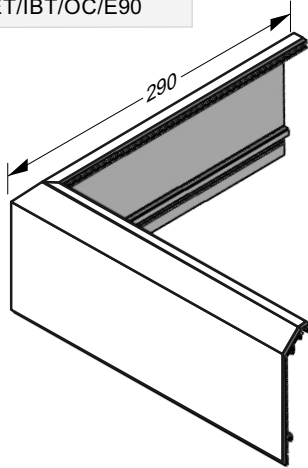
12mm Toughened	Use 2 x 2mm Gaskets
13.52mm SentryGlas	
15mm Toughened	Use 2 x 1mm Gaskets
15.2mm Laminated	
17.2mm Laminated	
17.52mm SentryGlas	

Gasket 2mm Thick	Front 100 wide, Part No JET/IB/GGF2 Rear 150 wide, Part No JET/IB/GGR2
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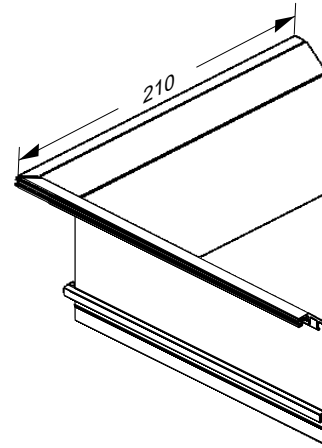
Gasket 1mm Thick	Front 100 wide, Part No JET/IB/GGF1 Rear 150 wide, Part No JET/IB/GGR1
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Juralco Edgetec Infinity® Balustrade System
Top Fix Components

Top Mount External Corner Cover
Part No JET/IBT/OC/E90



Top Mount Internal Corner Cover
Part No JET/IBT/OC/I90



Juralco Edgetec Infinity® Balustrade System

Typical Fixing - Residential and Commercial

Typical TOP Fix to Timber, Single Joist - M12 SS Coachscrew

Complies with NZS3604:2011 - Single Boundary Joist

Up to and including Very High Wind Zone Residential A, A Other and C3 only			Up to and including Extra High Wind Zone Residential A, A Other and C3 only			Up to and including Extra High Wind Zone Commercial B, E and C3 only		
Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)	Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)	Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)
12 T	1200	500	12 T	1000	500	15 T	1300	400
15.2 L	1150	500				17.2 L	1300	400
13.52SG	1050	350				17.52SG	1200	350

Height/Spacings for this mounting type only

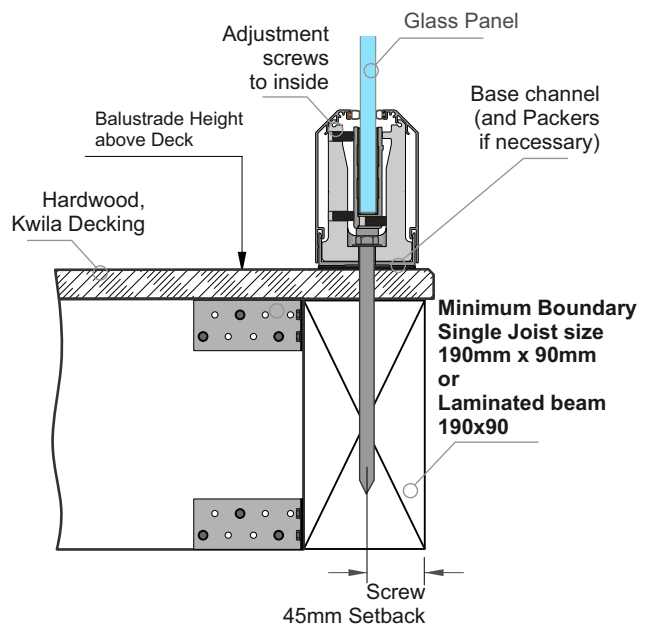
Up to and including Very High Wind Zone Pool Fence only			Up to and including Extra High Wind Zone Pool Fence only		
Glass Thickness, Type	Fence Height (max)	Clamp Spacing (max)	Glass Thickness, Type	Fence Height (max)	Clamp Spacing (max)
12T	1250	500	15T	1250	500

Applies to Pool Fences not protecting a fall of 1.0m or more

Glass Thickness, Type	Fence Height (max)	Clamp Spacing (max)	Glass Thickness, Type	Fence Height (max)	Clamp Spacing (max)
12T	1250	500	15T	1250	500

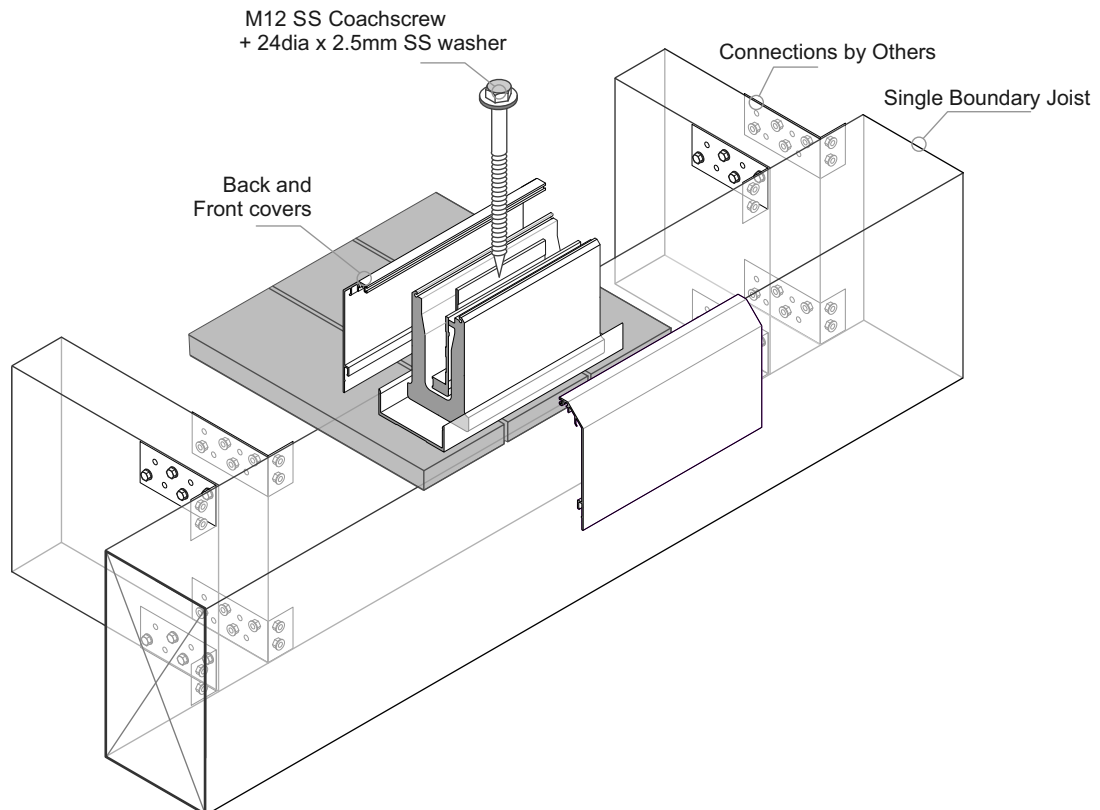
General Notes:

- 1 - Glass thickness, mm
Glass type T= Toughened, L = Laminated, SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to Elevations for Max Panel widths.
Use of Top Interlinking Rails (T and L only) or Stiffener Brackets (L and SG only)



Important Installation notes:

- 1 - The Project Engineer must ensure the structure can support the appropriate loads
- 2 - Substructure shown indicatively only. Timber SG8 minimum strength
- 3 - Coachscrews 150mm min engagement into joists, predrill 6mm holes.
- 4 - Bond all coachscrews with SIKa Supergrip to full depth
- 5 - All Fixings must be Stainless steel



Juralco Edgetec Infinity® Balustrade System

Typical Fixing - Residential and Commercial

Typical TOP Fix to Timber, Triple Joist - M12 SS Coachscrew

Complies with NZS3604:2011 - Triple Boundary Joist

**Up to and including
Very High Wind Zone**
Residential
A, A Other and C3 only

Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)
12 T	1200	500
15.2 L	1150	500
13.52SG	1050	350

**Up to and including
Extra High Wind Zone**
Residential
A, A Other and C3 only

Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)
12 T	1000	500

**Up to and including
Extra High Wind Zone**
Commercial
B, E and C3 only

Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)
15 T	1300	400
17.2 L	1300	400
17.52SG	1200	350

Height/Spacings for this mounting type only

**Up to and including
Very High Wind Zone**
Pool Fence only

Glass Thickness, Type	Fence Height (max)	Clamp Spacing (max)
12T	1250	500

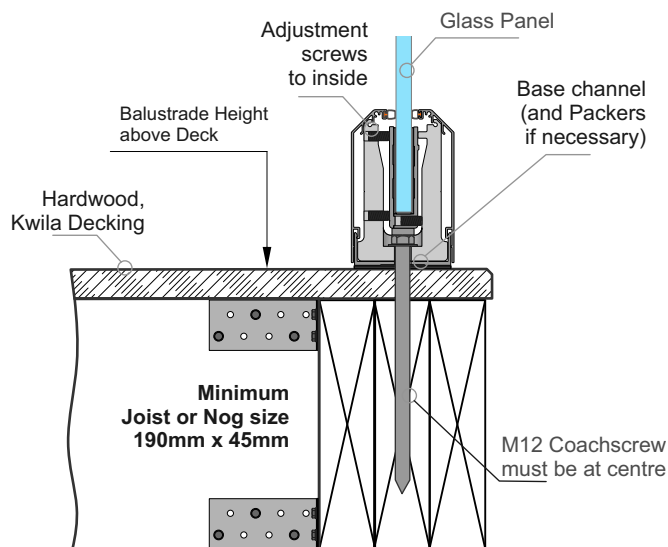
**Up to and including
Extra High Wind Zone**
Pool Fence only

Glass Thickness, Type	Fence Height (max)	Clamp Spacing (max)
15T	1250	500

Applies to Pool Fences not protecting a fall of 1.0m or more

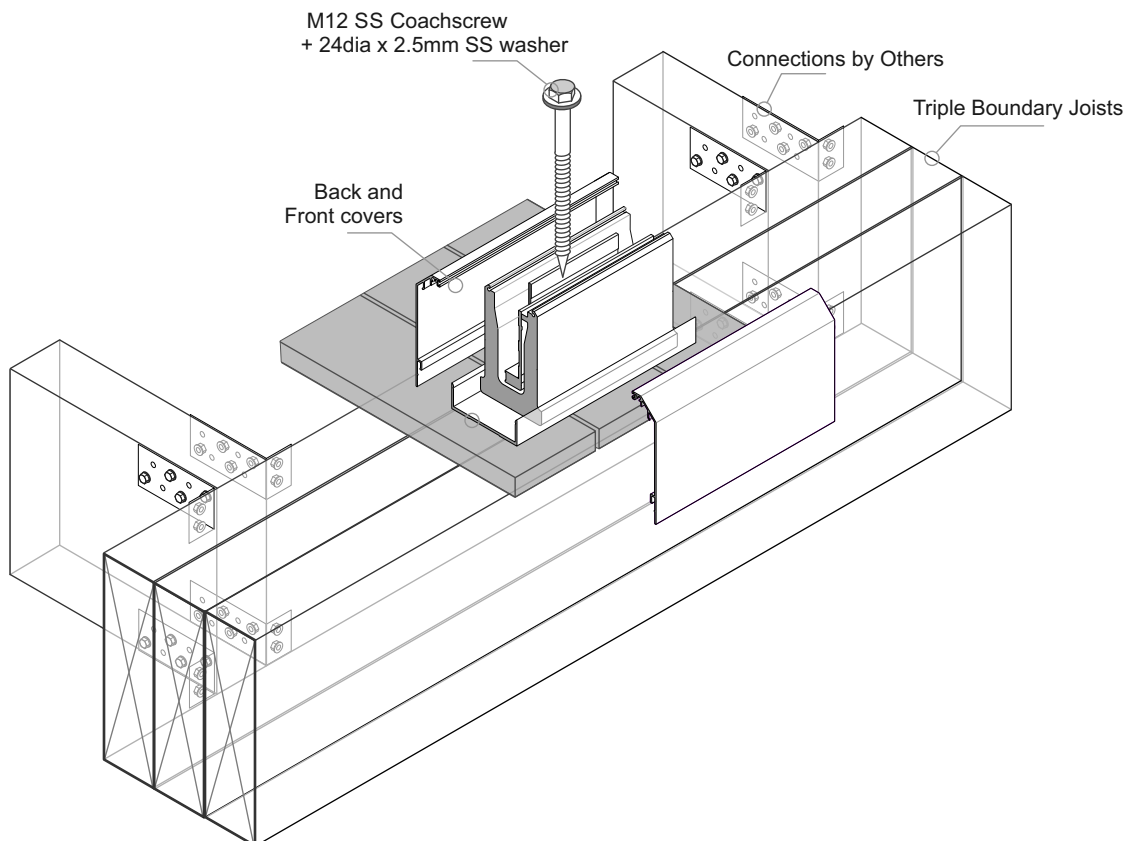
General Notes:

- 1 - Glass thickness, mm
Glass type T= Toughened, L = Laminated, SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to Elevations for Max Panel widths.
Use of Top Interlinking Rails (T and L only) or Stiffener Brackets (L and SG only)



Important Installation notes:

- 1 - The Project Engineer must ensure the structure can support the appropriate loads
- 2 - Substructure shown indicatively only. Timber SG8 minimum strength
- 3 - Coachscrews 150mm min engagement into joists, predrill 6mm holes.
- 4 - Bond all coachscrews with SIKa Supergrip to full depth
- 5 - All Fixings must be Stainless steel



Juralco Edgetec Infinity® Balustrade System

Typical Fixing - Residential and Commercial

Typical TOP Fix to Steel + Timber Deck - M12 SS, Bolt or Threaded Rod

**Up to and including
Very High Wind Zone
Residential
A, A Other and C3 only**

Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)
12 T	1200	500
15.2 L	1150	500
13.52SG	1050	350

**Up to and including
Extra High Wind Zone
Residential
A, A Other and C3 only**

Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)
12 T	1000	500

**Up to and including
Extra High Wind Zone
Commercial
B, E and C3 only**

Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)
15 T	1300	400
17.2 L	1300	400
17.52SG	1200	350

Height/Spacings for this mounting type only

**Up to and including
Very High Wind Zone
Pool Fence only**

Glass Thickness, Type	Fence Height (max)	Clamp Spacing (max)
12T	1250	500

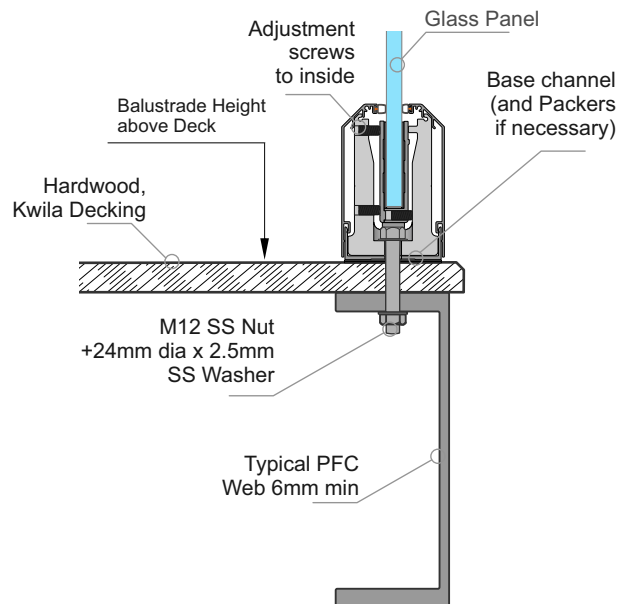
**Up to and including
Extra High Wind Zone
Pool Fence only**

Glass Thickness, Type	Fence Height (max)	Clamp Spacing (max)
15T	1250	500

Applies to Pool Fences not protecting a fall of 1.0m or more

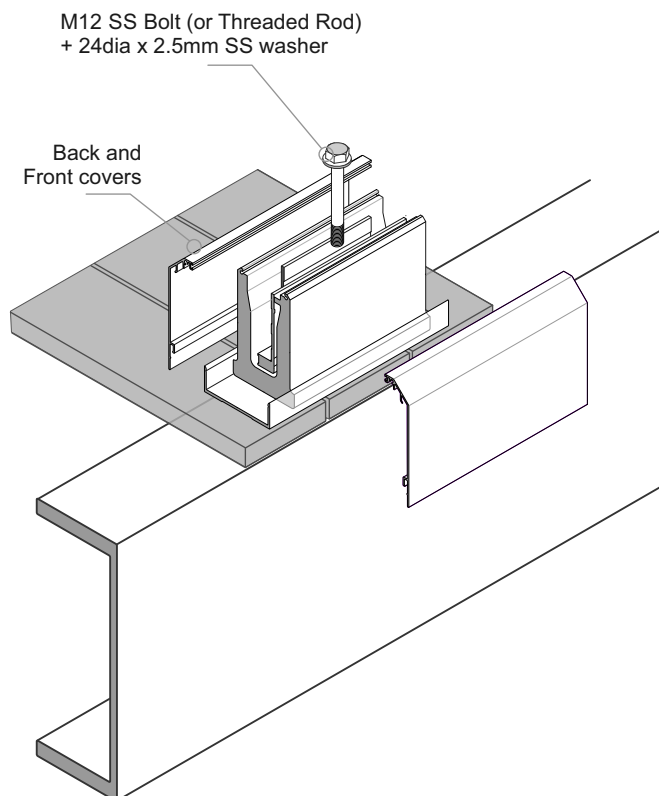
General Notes:

- 1 - Glass thickness, mm
Glass type T= Toughened, L = Laminated, SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to Elevations for Max Panel widths.
Use of Top Interlinking Rails (T and L only) or Stiffener Brackets (L and SG only)



Important Installation notes:

- 1 - The Project Engineer must ensure the structure can support the appropriate loads
- 2 - Substructure shown indicatively only
- 3 - All fixings must be Stainless Steel



Juralco Edgetec Infinity® Balustrade System

Typical Fixing - Residential and Commercial

Typical TOP Fix directly to Steel - M12 SS, Bolt or Threaded Rod

**Up to and including
Very High Wind Zone
Residential
A, A Other and C3 only**

Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)
12 T	1200	500
15.2 L	1150	500
13.52SG	1050	350

**Up to and including
Extra High Wind Zone
Residential
A, A Other and C3 only**

Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)
12 T	1000	500

**Up to and including
Extra High Wind Zone
Commercial
B, E and C3 only**

Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)
15 T	1300	400
17.2 L	1300	400
17.52SG	1200	350

Height/Spacings for this mounting type only

**Up to and including
Very High Wind Zone
Pool Fence only**

Glass Thickness, Type	Fence Height (max)	Clamp Spacing (max)
12T	1250	500

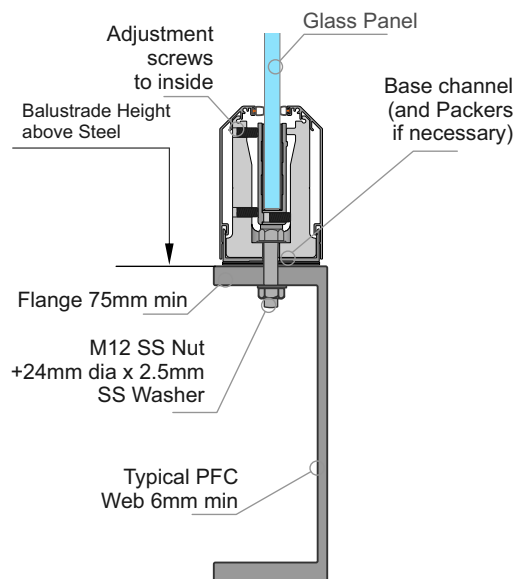
**Up to and including
Extra High Wind Zone
Pool Fence only**

Glass Thickness, Type	Fence Height (max)	Clamp Spacing (max)
15T	1250	500

Applies to Pool Fences not protecting a fall of 1.0m or more

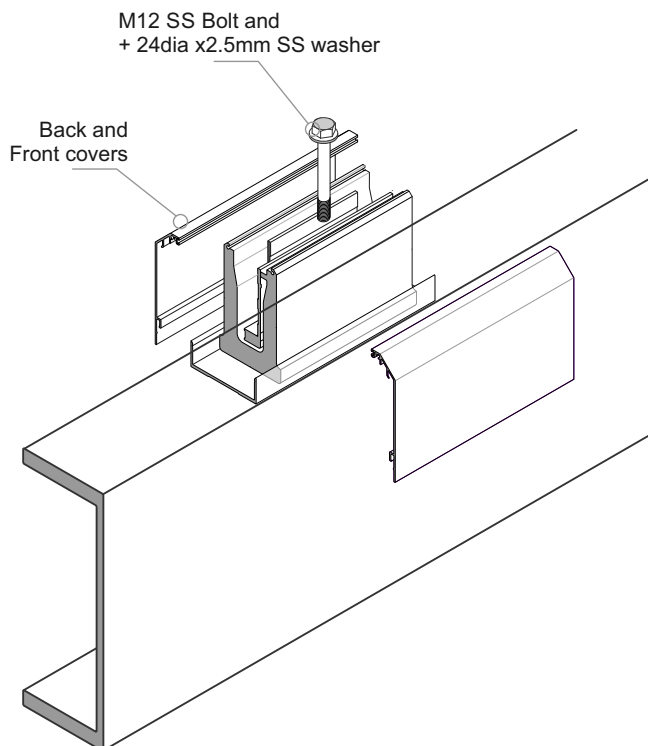
General Notes:

- 1 - Glass thickness, mm
Glass type T= Toughened, L = Laminated, SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to Elevations for Max Panel widths.
Use of Top Interlinking Rails (T and L only) or Stiffener Brackets (L and SG only)



Important Installation notes:

- 1 - The Project Engineer must ensure the structure can support the appropriate loads
- 2 - An PVC Tape layer must be installed between the Base Channel and Steel
- 3 - All fixings must be Stainless Steel



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Juralco Edgetec Infinity® Balustrade System

Issue 2-25 v1
Page 37

Juralco Edgetec Infinity® Balustrade System

Typical Fixing - Residential and Commercial

Typical TOP Fix to Concrete - M12 SS Threaded Rod Stud

Up to and including Very High Wind Zone Residential A, A Other and C3 only			Up to and including Extra High Wind Zone Commercial B, E and C3 only		
Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)	Glass Thickness, Type	Balustrade Height (max)	Clamp Spacing (max)

12 T	1200	500	15 T	1300	400
15.2 L	1150	500	17.2 L	1300	400
13.52SG	1050	350	17.52SG	1200	350

Height/Spacings for this mounting type only

Up to and including Very High Wind Zone Pool Fence only			Up to and including Extra High Wind Zone Pool Fence only		
Glass Thickness, Type	Fence Height (max)	Clamp Spacing (max)	Glass Thickness, Type	Fence Height (max)	Clamp Spacing (max)

Applies to Pool Fences not protecting a fall of 1.0m or more

12T	1250	500	15T	1250	500
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General Notes:

- 1 - Glass thickness, mm
Glass type T= Toughened, L = Laminated, SG = SentryGlas
- 2 - All measurements mm
- 3 - Refer to Elevations for Max Panel widths.
Use of Top Interlinking Rails (T and L only) or Stiffener Brackets (L and SG only)

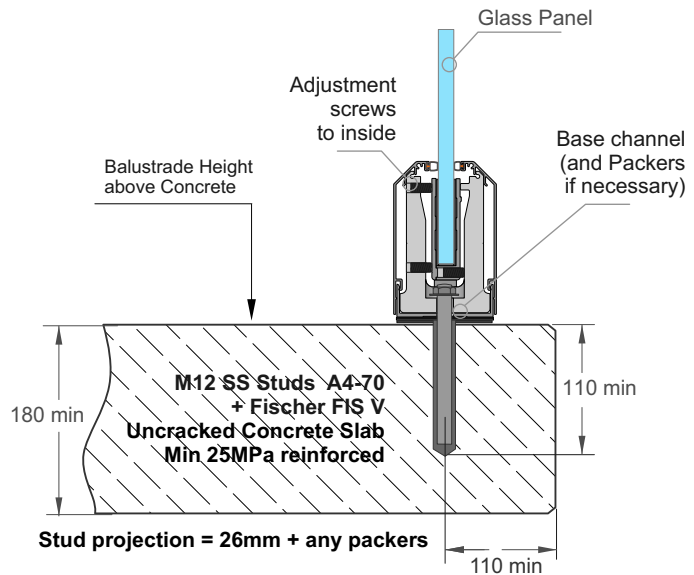


Installation details Fischer FIS V 300T

Thread diameter	M12
Drill hole diameter	= 14 mm
Drill hole depth	= 120 mm
Anchorage depth	= 110 mm

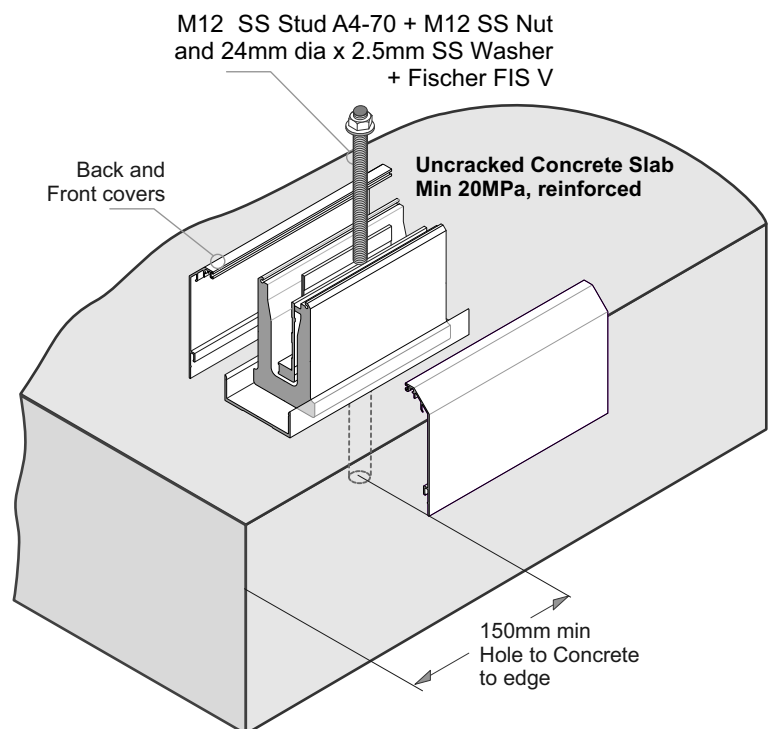
Drilling method	Hammer drilling
Drill hole cleaning	4 times blowing, 4 times brushing, 4 times blowing

No borehole cleaning required in case of using a hollow drill bit, e.g. fischer FHD.



Important Installation Notes:

- 1 - The Project Engineer must ensure the structure can support the appropriate loads
- 2 - Substructure shown indicatively only
- 3 - Fixings must engage into the structural slab
- 4 - A PVC Tape layer must be installed between the Base Channel and Concrete
- 5 - Use Threadlok on Nut
- 6 - All fixings must be Stainless Steel



Juralco Edgetec Infinity® Balustrade System
Top Fix Installation Recommendations

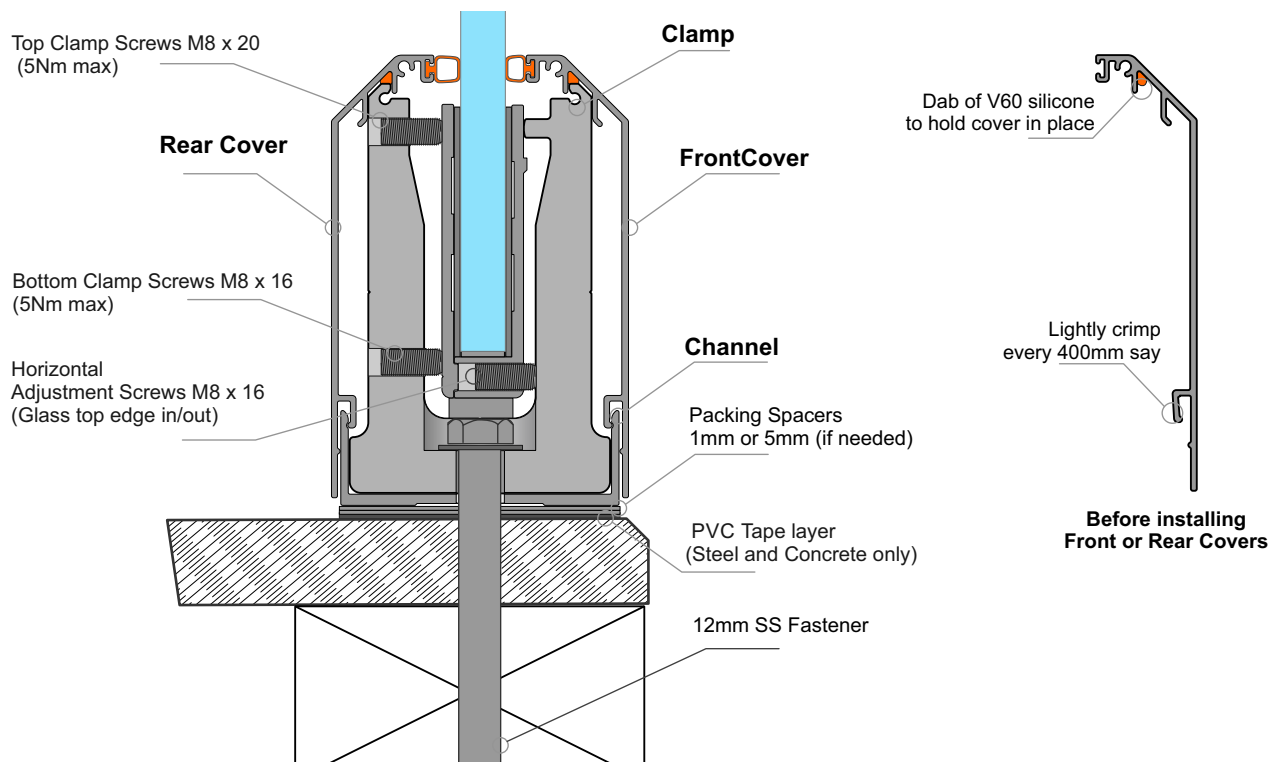
These Top Fix Installation Recommendations apply to all Substrates

Infinity Balustrade Top Fix Installation procedure on a Timber Deck. The Clamps must be Plumb and in Line

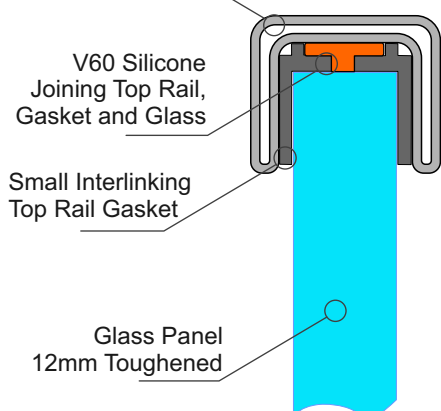
- 1 - Place Channel on deck. Lightly screw in position with 8g SS C/s screws, in a straight line.
 - 2 - Measure vertical height to deck at ea attach point. Calculate spacers needed to bring channel level (1mm and 5mm)
 - 3 - Set heights with spacers, including a PVC Tape layer to Deck (Steel and Concrete).
 - Tighten 8g C/s SS screws to firmly locate channel on deck. Channel should now be firm, level and straight
 - 4 - Mark out position of Clamps (normally at Glass Panel joints). Pre drill Deck for appropriate fastener, through channel.
 - 5 - Place Clamps in place (adjust screws to inside). Fasten all down very firmly. **Clamps must be Plumb and in Line**
 - 6 - Place Front cover in place, incl bulb seal. Lightly crimp bottom tag, and a dab of V60 silicone at top.
-
- 7 - Fit Glass into position inside the Clamps, using appropriate Gasket combinations.
 - As the channel is already level, there are no provision for Glass vertical adjustment
 - 8 - Lightly nip the top 2 grub screws on the HD clamp to hold the glass vertical.
 - 9 - Adjust the 4 lower grub screws on the HD clamp and Glass clamp assemblies for top edge Horizontal alignment
 - 10 - When glass panels are in the correct position tighten top and bottom clamp screws on HD clamp (5.0Nm max)
-
- 11 - Install Rear cover incl bulb seal. Lightly crimp bottom tag, and a dab of V60 silicone at top.
 - 12 - Fit End plates as required

Fitting Stages 1 - 6 to get Clamp Plumb, both Vertical and Horizontal

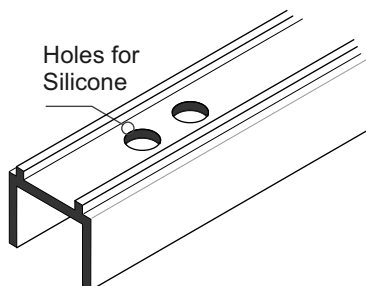
Fitting Stages 7 - 10 to get Glass Plumb, both Vertical and Horizontal



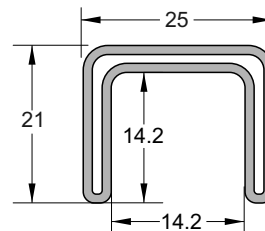
Small SS Interlinking Top Rail



25mm SS Interlinking Top Rail



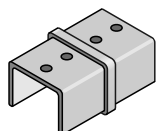
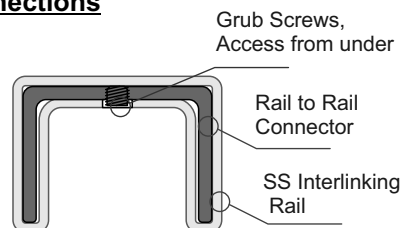
SMALL SS INTERLINKING TOP RAIL GASKET
JET/490GT/12/2.9 (Black)



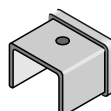
SMALL SS INTERLINKING TOP RAIL
JET/490/5.8/SSS JET/490/5.8/SCC
Duplex 2205

25mm SS Interlinking Rail Connections

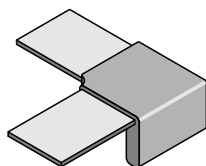
Note : All these Brackets use M5 x 6 SS Grub Screws.
If necessary these holes must be Drilled + tapped M5, as shown.
The under side of the Interlinking Rail must be drilled M6 to match M5 tapped holes positions, for access to Grub screws
- Joins, where required must be at the end of Glass Panels
Available as Satin(SSS) or Powdercoated SCC finishes



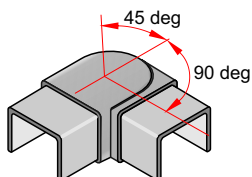
180deg INLINE JOINER
Duplex 2205
JET491/SSS JET491/SCC
21mm x 25mm x 51mm deep



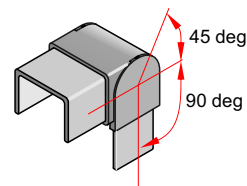
END CAP Duplex 2205
JET492/SSS JET492/SCC
21mm x 25mm x 25mm deep



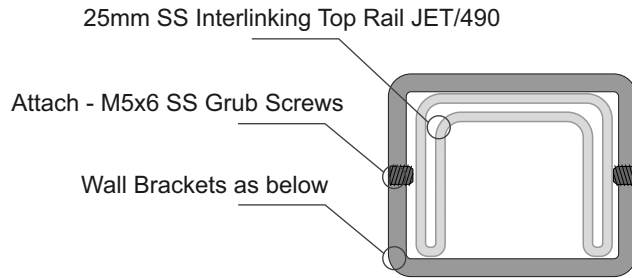
90deg JOINER Duplex 2205
JET493/SSS JET493/SCC
21mm x 80mm x 80mm



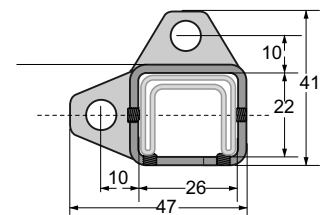
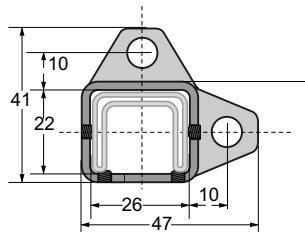
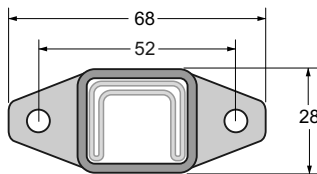
+90 to - 45 deg ADJUSTABLE HORIZONTAL JOINER Duplex 2205
JET494/SSS JET494/SCC
21mm x 25mm x 75mm overall deep



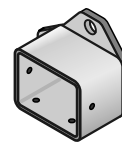
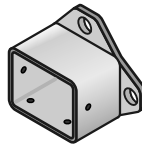
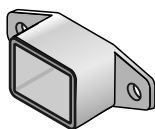
+90 to - 45 deg ADJUSTABLE VERTICAL JOINER Duplex 2205
JET495/SSS JET495/SCC
21mm x 25mm x 73mm overall deep



Brackets for Fixing to Wall or End Post for 25mm SS Interlinking Rail



Note : All these Brackets use M5x6mm SS Grub Screws



WALL BRACKET Duplex 2205
JET496/SSS JET/496/SCC
68mm x 28mm x 30mm deep

WALL BRACKET - RH Duplex 2205
JET497/RH/SSS JET497/RH/SCC
41mm x 47mm x 30mm deep

WALL BRACKET - LH Duplex 2205
JET497/LH/SSS JET497/RH/SCC
41mm x 47mm x 30mm deep

General Notes:

- All fixings to be Stainless Steel. - PVC Tape layer between Structure and Bracket
- ULS Point load $N^* = 0.9\text{kN}$, inwards, outwards or down and in tension

Note : Fixing to Steel

- use 2 off 8g SS TEK Screws or M6 SS Bolts
- Steel 2mm min thickness
- Steel 300MPa minimum
- 15mm min distance to any Edges

Note : Fixing to Timber Wall

- use 2 off 8g SS Screws, 35mm min into studs.
- use Sika Supergrip 2hr
- 30mm min distance to Horizontal Edge
- If Weatherboard use suitable predrilled Wedge
- Timber stud wall to be designed and detailed in accordance with NZ3603 or NZ3604

Note : Fixing to Juralco EDGE Post

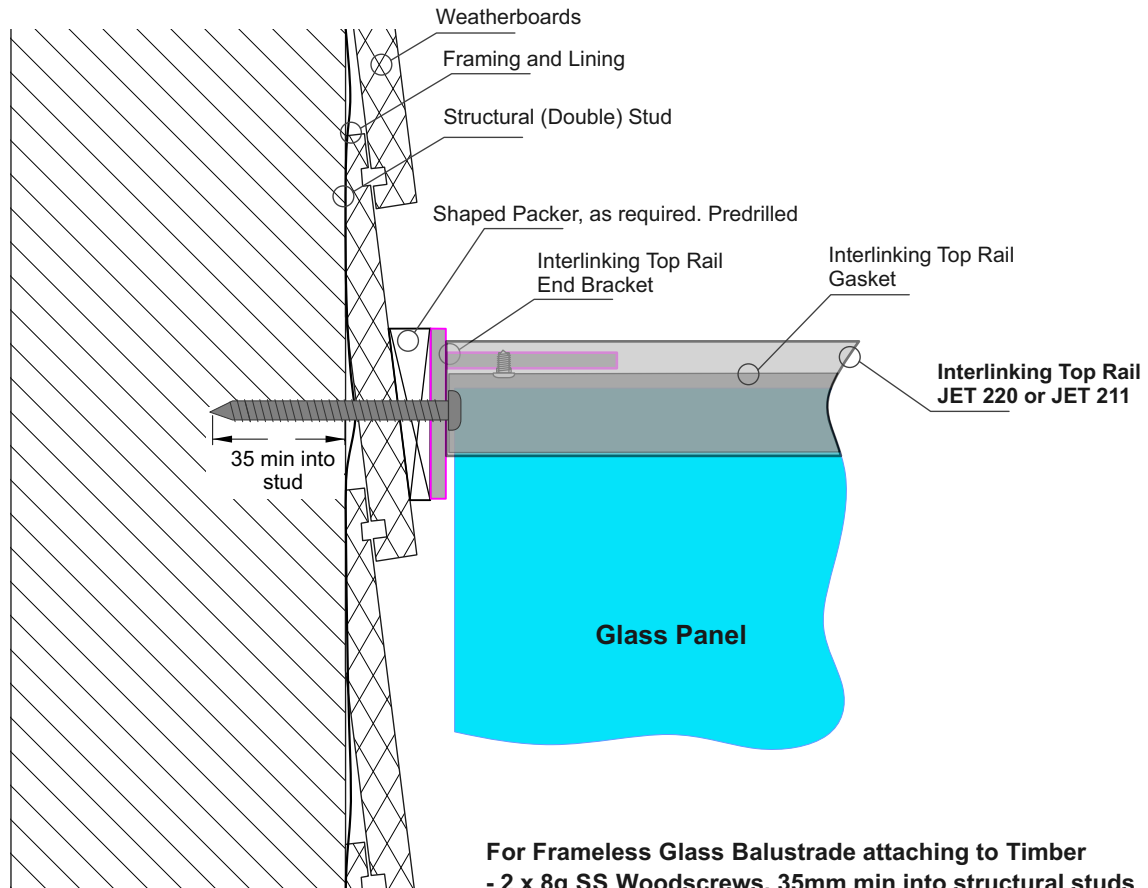
- use 2 off 8g x 25 SS PK Screws

Note : Fixing to Concrete Wall

- use 2 off M6 x70 SS Screw Anchors
- Solid Concrete min 20Mpa
- Block wall Concrete filled/Reinforced
- 140mm min Wall thickness
- 70mm min distance to Horizontal Edge
- 100mm min distance to Vertical Edge
- Blockwork wall must be corefilled /reinforced and is to be designed and detailed in accordance with NZ4230 or NZ4229

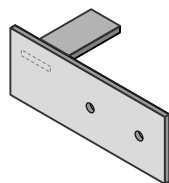
Important Note: All Interlinking rails, at their ends must be attached to a Building Structure or to an Edge Post attached to the Deck structure, using Rail End Plates/Brackets



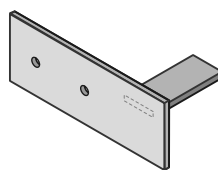


For Frameless Glass Balustrade attaching to Timber
 - 2 x 8g SS Woodscrews, 35mm min into structural studs
 - 20mm min to edge of stud in all directions

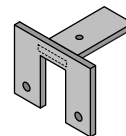
Interlinking Top Rail End Bracket Options - Drawing above shows JET40



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
Part No JET 40LH



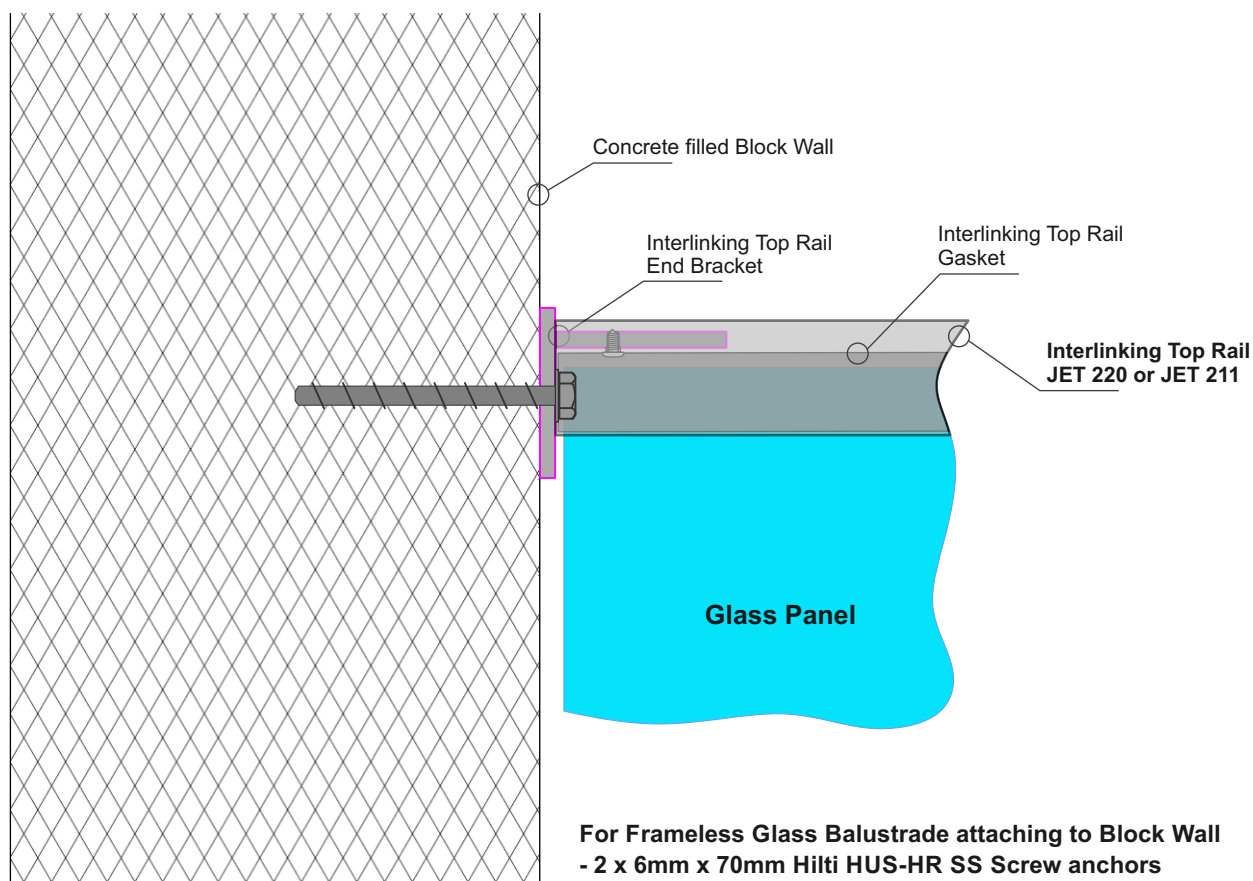
Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
Part No JET 40RH



Interlinking Top Rail
End Bracket
SS. 60mm x 46mm
Part No JET 42

Notes:

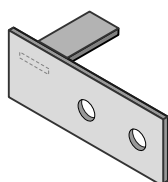
- All fixings to be stainless steel
- Timber stud wall to be designed by Project structural engineer for loads imposed by Balustrade.
- ULS Point load $N^* = 0.9\text{kN}$, inwards, outwards or down.
- Minimum Stud size = 90mm x 45mm
- Minimum Timber grade = Sg8
- Timber stud wall to be designed and detailed in accordance with NZ3603 or NZ3604



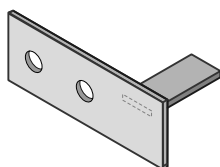
For Frameless Glass Balustrade attaching to Block Wall

- 2 x 6mm x 70mm Hilti HUS-HR SS Screw anchors
- For concrete drill 6mmØ holes
- 70mm min to side edge of concrete, 100mm to top edge.

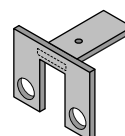
Interlinking Top Rail End Bracket Options - Drawing above shows JET40



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
Part No JET 40LH



Interlinking Top Rail
Wall type End Plate
SS. 120x45mm
Part No JET 40RH



Interlinking Top Rail
End Bracket
SS. 60mm x 46mm
Part No JET 42

Drill out holes to 9mmØ

Notes:

- All fixings to be stainless steel
- Blockwall to be designed by Project structural engineer for loads imposed by Balustrade.
- ULS Point load $N^* = 0.9\text{kN}$, inwards, outwards or down.
- Minimum blockwork thickness = 140mm
- Minimum core fill concrete strength = 17.5MPa
- Blockwork wall must be corefilled /reinforced and is to be designed and detailed in accordance with NZ4230 or NZ4229